

Soda lime zirconia silicate glasses as prospective hosts for zirconia-containing radioactive wastes

John G. Fisher^{*}, Peter F. James, John M. Parker

Department of Engineering Materials, University of Sheffield, Sir Robert Hadfield Building, Mappin Street, Sheffield S1 3JD, United Kingdom

Received 16 July 2004; received in revised form 28 January 2005

Abstract

$\text{Na}_2\text{O}-\text{CaO}-\text{ZrO}_2-\text{SiO}_2$ glass compositions with ZrO_2 contents of up to 20 mol% were melted. Up to 12.3 mol% ZrO_2 could be dissolved into the glasses. Melting temperatures $\geq 1450^\circ\text{C}$ were required to remove seed and produce a melt that could be cast. Addition of ZrO_2 caused an increase in the glass transition and crystallization temperatures. Glasses crystallized at temperatures $\geq 1050^\circ\text{C}$ with Keldyshite and Parakeldyshite ($\text{Na}_2\text{O} \cdot \text{ZrO}_2 \cdot 2\text{SiO}_2$) as the crystalline phases. Addition of up to 4.6 mol% ZrO_2 caused an increase in the hydrolytic resistance of the glass, with further additions having little effect. The suitability of these glasses as hosts for ZrO_2 -containing radioactive wastes is discussed.

© 2005 Published by Elsevier B.V.

PACS: 81.05.K

1. Introduction

There is a need for glasses that can immobilize radioactive wastes containing high levels of Zr [1]. This Zr comes from the cladding materials used to enclose the UO_2 fuel in the fuel cans. For example, in pressurized water reactors (PWR) and Canadian deuterium uranium reactors (CANDU), the UO_2 fuel is clad in zirconium alloy fuel cans. In Soviet-designed reactors the fuel is clad in zirconium–niobium alloy cans. When these fuel cans are reprocessed the cladding is stripped off and forms a separate radioactive waste stream [2]. In addition, during reprocessing of spent fuel a high level liquid waste product (HLLW) containing Zr leached from the fuel cans is produced [1]. High-zirconia borosilicate

glasses with ZrO_2 contents of up to 25 wt% have been studied as potential candidates for immobilization of Zr-containing waste [2–4]. However, these glasses require melting temperatures of up to 1450°C . In addition to borosilicate glasses, zirconia-containing silicate glasses with ZrO_2 contents of up to 23.5 wt% have also been studied [5].

Suitable waste compositions would require the following characteristics [6,2]:

1. high chemical durability towards aqueous attack;
2. melting temperatures of $\leq 1150^\circ\text{C}$, in order to reduce loss of radioactive volatiles such as Na or Ce and permit melting in induction or joule-heated nuclear waste vitrification processes;
3. the ability to accommodate appreciable amounts of ZrO_2 ;
4. high glass transition temperature so that the glass is kinetically stable with respect to crystallization under storage conditions;
5. homogeneity of the glass.

^{*} Corresponding author. Present address: Department of Materials Science and Engineering, Korea Advanced Institute of Science and Technology, 373-1 Kusong-dong, Yusong-gu, Daejeon 305-701, Korea. Tel.: +82 42 869 4153; fax: +82 42 869 8920.

E-mail address: jgfisher@kaist.ac.kr (J.G. Fisher).

Relatively little work has been done on producing glasses in the Na_2O – CaO – ZrO_2 – SiO_2 system. Dumbleby et al. [7] melted Na_2O – CaO – ZrO_2 – SiO_2 glasses of the following compositions: 17.5–20.9 mol% Na_2O –3.75 mol% CaO –0.3–3.75 mol% ZrO_2 –75 mol% SiO_2 . Batches were melted at 1350 °C for 2 h to give clear glasses. Trying to introduce more ZrO_2 under these conditions resulted in opaque glasses due to the presence of undissolved ZrO_2 .

Navarro et al. [8] melted batches of composition 16 mol% Na_2O –10 mol% CaO – x mol% ZrO_2 –(74– x) mol% SiO_2 , where $x \leq 21$. Compositions were melted at temperatures between 1500 and 1600 °C. Monoclinic ZrO_2 crystallized on cooling from compositions containing ≥ 15 mol% ZrO_2 . The glasses showed phase separation when examined by transmission electron microscopy. Hydrolytic resistance, measured as the amount of Na_2O extracted from the glass, was found to increase with increasing ZrO_2 content. Comparison of extraction rates for SiO_2 and ZrO_2 showed that a ZrO_2 -rich layer formed on the surface of the leached glass. Ohta and Suzuki [9] studied the effect of substituting ZrO_2 for silica in a base glass of composition 15 wt% Na_2O –12 wt% CaO –73 wt% SiO_2 . 25 wt% of ZrO_2 could be substituted for SiO_2 . Addition of ZrO_2 was found to improve the hydrolytic durability of the base glass, measured as the amount of Na_2O extracted from the glass.

Glasses in the Na_2O – CaO – ZrO_2 – SiO_2 system have not previously been studied with respect to their suitability as waste storage glasses. Therefore, several compositions from this system were studied to this aim. The Na_2O – CaO – SiO_2 composition 21.4 mol% Na_2O –4.1 mol% CaO –74.5 mol% SiO_2 was chosen as a base composition for the Na_2O – CaO – ZrO_2 – SiO_2 glasses. This is a low temperature (776 °C) eutectic in the Na_2O – CaO – SiO_2 ternary system [10]. By increasing the amount of ZrO_2 added to the base composition, the liquidus temperature of the composition can be increased.

2. Experimental

Compositions containing 0, 5, 7.5, 10, 12.5, 15, 17.5 and 20 mol% ZrO_2 were melted. Na_2CO_3 (Fisher SLR Grade), CaCO_3 (Fisher SLR Grade), SiO_2 (Tilcon L60A Loch Aline) and ZrO_2 (Avocado SLR Grade) were used as batch materials. Compositions are shown in Table 1. The $\text{Na}_2\text{O}/\text{CaO}/\text{SiO}_2$ ratio of the glasses was kept constant. The calculated compositions and the actual compositions as determined by Inductively Coupled Plasma Optical Emission Spectroscopy are given.

Compositions P0–P3 were melted in an electric furnace at 1450 °C. Compositions P4–P7 were melted in a gas furnace at 1620–1650 °C. Melting time was 6 h in

Table 1
Composition of Na_2O – CaO – ZrO_2 – SiO_2 glasses

Glass	Oxide	Mol% calculated	Mol% analyzed
P0	Na_2O	21.41	–
	CaO	4.1	–
	ZrO_2	0	–
	SiO_2	74.49	–
P1	Na_2O	20.34	18.48
	CaO	3.9	3.88
	ZrO_2	5	4.59
	SiO_2	70.77	73.05
P2	Na_2O	19.8	19.13
	CaO	3.79	3.69
	ZrO_2	7.5	6.28
	SiO_2	68.9	70.90
P3	Na_2O	19.27	15.32
	CaO	3.69	3.17
	ZrO_2	10	7.72
	SiO_2	67.04	73.78
P4	Na_2O	18.73	17.63
	CaO	3.59	3.68
	ZrO_2	12.5	11.09
	SiO_2	65.18	67.60
P5	Na_2O	18.2	17.32
	CaO	3.49	3.39
	ZrO_2	15	12.28
	SiO_2	63.32	67.00
P6	Na_2O	17.66	17.40
	CaO	3.38	3.32
	ZrO_2	17.5	14.72
	SiO_2	61.45	64.55
P7	Na_2O	17.13	17.49
	CaO	3.28	3.53
	ZrO_2	20	15.25
	SiO_2	59.59	63.73

each case. Melting took place in Pt crucibles. The melted compositions were cast into blocks and annealed at 500 °C for 1 h, then cooled at 1 °C min^{−1} to room temperature.

Chemical analysis was carried out on the glasses by Inductively Coupled Plasma Optical Emission Spectroscopy. X-ray diffraction (XRD) was carried out on powder samples in a Phillips diffractometer (step size 0.02°, scan speed 2° min^{−1}). Differential thermal analysis (DTA) was carried out using Perkin Elmer equipment with a heating rate of 10 °C min^{−1} and a maximum temperature of 1000–1300 °C. The extrapolated onset of peaks was measured.

To determine the crystallization behavior, samples from all the glasses were heated for 2 h at 800 °C, 850 °C, 900 °C and 950 °C, 12 h at 950 °C, 12 h at 1050 °C and 12 h at 1150 °C. In addition, samples from glasses P1A and P2 were heated for 2 h at 750 °C. Samples were heated in a Lenton muffle oven calibrated to ± 3 °C. Heating and cooling rates for all the samples were 1 °C min^{−1}. Samples for optical microscopy were

polished to a 1 μm finish and then etched for 30 s in a 5 vol.% HF solution.

Chemical durability tests were carried out on solid samples of compositions P1–P5 using the Soxhlet leach test. Samples from the glasses were cut and ground into cuboids with perpendicular edges. Samples were then polished to a 1 μm finish using diamond paste. Paraffin and an oil-based lubricant were used during the grinding and polishing stages. The samples were cleaned and the surface area and weight measured. The samples were then leached in distilled water at room temperature for 28 days. The samples were reweighed and the bulk leach rate calculated from the formula:

$$\text{Bulk leach rate (g m}^{-2} \text{ day}^{-1}) = \frac{\text{Sample weight loss (g)}}{\text{Surface area (m}^2) \times \text{length of test (days)}}. \quad (1)$$

3. Results

All compositions except P6 and P7 (17.5 and 20 mol% ZrO_2 as calculated) produced clear glasses which appeared free of devitrification to the naked eye. Compositions P6 and P7 produced clear glasses which were partially crystalline. When composition P4 (12.5 mol% ZrO_2 as calculated) was melted, some undissolved batch remained in the crucible after pouring. Wang et al. [5] in their study of ZrO_2 -containing glasses, found undissolved batch remaining in the crucible after several melts, all of which were stirred. They concluded that this was due to the relatively dense ZrO_2 sinking to the bottom of the crucible where the action of the stirrer was less effective. On remelting compositions at lower temperatures (and hence having melts with higher viscosities) they found that more ZrO_2 could be incorporated into the glasses.

From the work done on glasses in the Na_2O – ZrO_2 – SiO_2 and Na_2O – CaO – ZrO_2 – SiO_2 systems by Wang et al. [5] and Navarro et al. [8], the liquidus temperatures of compositions P1–P7 were expected to be in the region of 1450–1600 $^\circ\text{C}$. This was confirmed by our experiments.

The compositions as determined by chemical analysis are given in Table 1. The maximum amount of ZrO_2 that could be incorporated into the glass was approximately 12.3 mol%.

3.1. Differential thermal analysis

A DTA trace for glass P5 is shown in Fig. 1. Small exothermic peaks are visible at 1059 $^\circ\text{C}$ and 1144 $^\circ\text{C}$. These are attributed to crystallization of the glasses. X-ray diffraction shows two crystalline phases, Keldyshite and Parakeldyshite, in the heat-treated glasses. An endothermic peak is visible at 1270 $^\circ\text{C}$. This may be

due to the sample melting or to a phase change. Exothermic crystallization peaks are also visible in the DTA trace for glass P4.

The values of T_g and the temperature of onset of crystallization, T_c , for the different compositions are shown in Table 2. The accuracy of the measurements is ± 4 $^\circ\text{C}$. No crystallization peaks could be determined in samples P0–P2. Samples from P3 and P6 were only heated to a maximum temperature of ~ 970 $^\circ\text{C}$ so it is not known whether crystallization peaks exist for these glasses at temperatures > 1000 $^\circ\text{C}$. A graph of T_g vs. ZrO_2 content is shown in Fig. 2. The straight line is a guide only. The graph shows that T_g increased as the ZrO_2 content of the glasses increased.

3.2. X-ray diffraction

X-ray diffraction traces for the as-melted glasses (i.e. before heat-treatment) are shown in Fig. 3. Values for ZrO_2 content refer to the ZrO_2 contents as found by chemical analysis. All the glasses appeared amorphous to X-rays except P7 (15.25 mol% ZrO_2 as analyzed). This glass showed traces of baddeleyite (monoclinic ZrO_2). Optical microscopy suggested that this baddeleyite was batch ZrO_2 that had not dissolved.

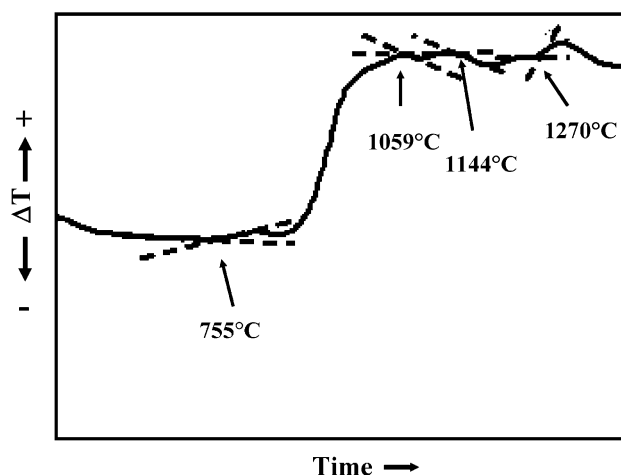
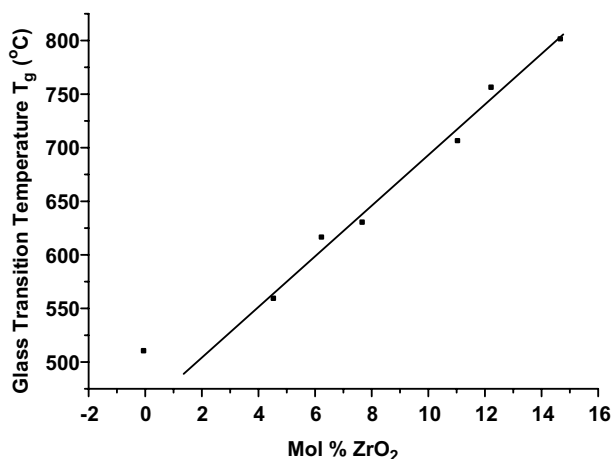
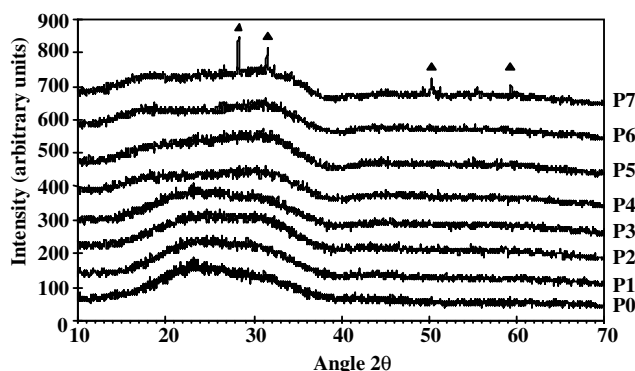


Fig. 1. DTA trace for glass P5.

Table 2

Values of T_g (glass transition temperature) and T_c (crystallization temperature)

Glass	Mol% ZrO_2 analyzed	T_g (± 4 $^\circ\text{C}$)	T_c (± 4 $^\circ\text{C}$)
P0	0	509 $^\circ\text{C}$	–
P1	4.59	558 $^\circ\text{C}$	–
P2	6.28	615 $^\circ\text{C}$	–
P3	7.72	629 $^\circ\text{C}$	–
P4	11.09	705 $^\circ\text{C}$	999 $^\circ\text{C}$, 1089 $^\circ\text{C}$
P5	12.28	755 $^\circ\text{C}$	1059 $^\circ\text{C}$, 1144 $^\circ\text{C}$
P6	14.72	800 $^\circ\text{C}$	–

Fig. 2. Graph of T_g vs. mol% ZrO_2 (analyzed).Fig. 3. X-ray diffraction traces for glasses P1–P7. (▲) Monoclinic ZrO_2 .

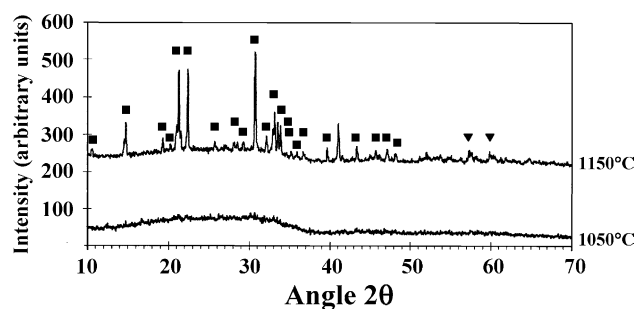
The XRD results for heat-treated glasses can be broken down into two groups: heat-treatments below 1050 °C and heat-treatments at and above 1050 °C. After heat-treatments below 1050 °C, most of the glasses appear completely amorphous to X-rays. Some of the glasses heat-treated below 1050 °C show crystallization peaks, but there are not enough peaks to determine the presence of compounds with any accuracy. Possible compounds are $CaZrO_3$, terskite ($2Na_2O \cdot ZrO_2 \cdot 6SiO_2$) and α -quartz.

The glasses began to devitrify when heat-treated at 1050 °C or above. The phases that crystallized during the different heat-treatments are given in Table 3. X-ray diffraction traces for glass P3 are shown in Fig. 4. Samples from glasses P1 and P2 melted and spheroidized when heat-treated at 1050 °C. They did not crystallize upon cooling. The fact that these two glasses spheroidized at 1050 °C meant that their liquidus temperature must have been between 950 °C and 1050 °C. Glasses P3–P5 have not crystallized after heat-treatment at 1050 °C, but in the XRD trace for glass P6, peaks for keldyshite and parakeldyshite are visible. After heat-

Table 3

Crystalline phases produced by heat-treatments at temperatures ≥ 1050 °C

Glass	Mol% ZrO_2 analyzed	1050 °C/12 h	1150 °C/12 h
P3	0	No phases	Keldyshite, $Na_2O \cdot ZrO_2 \cdot 2SiO_2$ Parakeldyshite, $Na_2O \cdot ZrO_2 \cdot 2SiO_2$
P4	4.59	No phases	Keldyshite Parakeldyshite
P5	6.28	No phases	Keldyshite Parakeldyshite
P6	7.72	Keldyshite Parakeldyshite	Keldyshite Parakeldyshite

Fig. 4. X-ray diffraction traces of glass P3 after heat-treatments at 1050 °C and 1150 °C respectively for 12 h. (■) Parakeldyshite ($Na_2O \cdot ZrO_2 \cdot 2SiO_2$); (▼) keldyshite ($Na_2O \cdot ZrO_2 \cdot 2SiO_2$).

treatment at 1150 °C, all of the glasses have crystallized into a mixture of keldyshite and parakeldyshite.

In the traces for glasses P3 and P4, amorphous ‘humps’ can still be seen at angles between 10° and 40°, indicating that only partial crystallization has taken place. In the traces for glasses P5 and P6, these ‘humps’ are no longer visible, indicating that complete crystallization has taken place. However, in the optical micrographs of these samples an amorphous phase is still present.

The relationship between keldyshite and parakeldyshite should be mentioned. Keldyshite and parakeldyshite have identical molecular formulae except that in keldyshite some of the Na ions are replaced by H. It was thought unlikely that any hydrogen would be present in crystals grown at ≥ 1050 °C and so in this paper the two phases have been given identical molecular formula.

3.3. Optical microscopy

As-melted (i.e. before heat-treatment) glasses P1–P5 appeared free of unmelted batch and devitrification. P6 appeared partially glassy, but some crystalline material was also present. An optical micrograph of this crystalline material is shown in Fig. 5. Spherical and oblate

particles 6–11 μm in diameter were clearly present. A few of the particles had angular corners. The light haze visible on the micrograph appeared to be due to more particles lying under the surface of the glass. The particles appeared to be more reflective than the surrounding glass, indicating that they have a higher refractive index than the surrounding glass [11]. In view of the shape and color of the particles, it is probable that they are unmelted ZrO_2 . P7 appeared very similar to P6 but more angular particles were visible, indicating that less dissolution of ZrO_2 had taken place. The size of the particles ranged from 3–13 μm . X-ray diffraction of this sample confirmed that monoclinic ZrO_2 was present in the glass.

Optical micrographs of samples P4 and P6 heat-treated at 1050 $^\circ\text{C}$ for 12 h are shown in Fig. 6. Devitrification has taken place at the surface. Two crystal phases were present: a lath-like phase and a dendritic phase. The lath-like crystals radiated outwards in fans from the surface of the sample, growing up to 380 μm into the sample. The dendritic phase appeared more reflective than the surrounding glass, indicating a higher refractive index than the surrounding glass. In glass P6, the dendritic phase had also crystallized in the bulk of the sample. White nodular crystals of unmelted ZrO_2 were also visible. Some of the crystals were porous. The parallel lines in the crystals were polishing scratches that had been preferentially etched.

Micrographs of samples P4 and P6 heat-treated at 1150 $^\circ\text{C}$ for 12 h are shown in Fig. 7. P4 has partially devitrified to give a lath-like crystalline phase in a glassy matrix. Crystallization appeared to have started in the bulk of the sample. Internal porosity could be seen within the crystals. These crystals appeared to be of the same phase as the crystals produced in heat-treatments at 1050 $^\circ\text{C}$ but did not radiate out in fans as the crystals produced in heat-treatments at 1050 $^\circ\text{C}$ did. X-ray diffraction showed that keldyshite and parakeldyshite were present so this lath-like crystalline phase must be keldyshite/parakeldyshite. The sample from glass P6 crys-

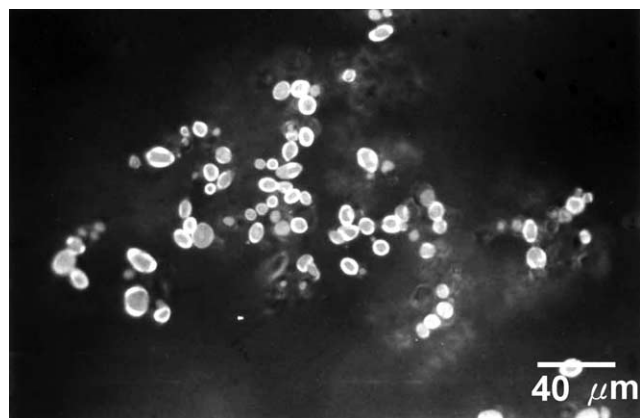


Fig. 5. Optical micrograph of glass P6 before heat-treatment.

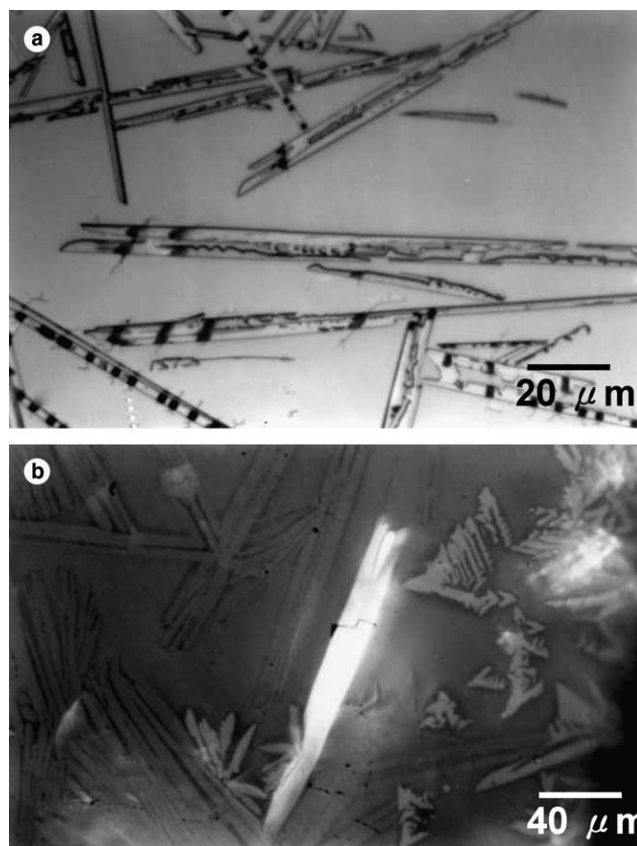


Fig. 6. Optical micrographs of (a) glass P4 and (b) glass P6 heat treated at 1050 $^\circ\text{C}$ for 12 h.

tallized into two phases throughout the sample: a dendritic phase and a minor phase of lath-like crystals. Crystallization appears to have started in the bulk of the sample. Both phases appeared more reflective than the surrounding amorphous matrix, suggesting that they have a higher refractive index. From the high reflectivity of the dendritic phase, it is thought that the dendritic phase is monoclinic ZrO_2 , although monoclinic ZrO_2 was absent from the XRD trace for this sample. X-ray diffraction showed that keldyshite and parakeldyshite were present. However, few lath-like crystals could be seen in this sample.

3.4. Chemical durability

During testing of all the samples, the samples remained intact and the leach water remained clear to the naked eye. The values of bulk leach rate are shown in Table 4. During testing of composition P4, the flask cracked and so this composition is not included in the results. A graph of bulk leach rate vs. mol% ZrO_2 (analyzed) is shown in Fig. 8. The bulk leach rate dropped rapidly with addition of 4.59 mol% ZrO_2 . Further additions of ZrO_2 caused little change in the bulk leach rate.

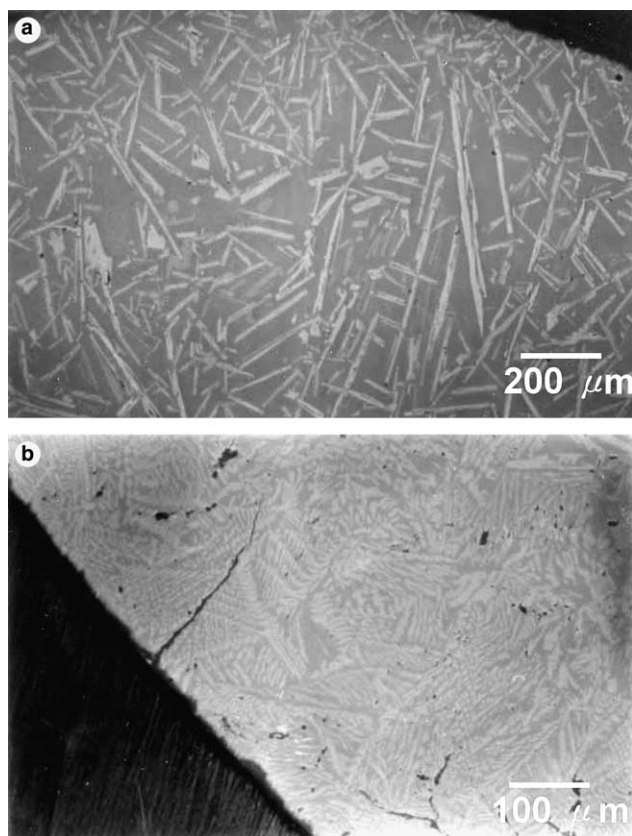


Fig. 7. Optical micrographs of (a) glass P4 and (b) glass P6 heat treated at 1150 °C for 12 h.

Table 4
Bulk leach rates of glasses

Glass	Mol% ZrO ₂ analyzed	Bulk leach rate (g m ⁻² day ⁻¹)
P0	0	8.89
P1	4.59	1.11×10^{-1}
P2	6.28	5.73×10^{-2}
P3	7.72	3.95×10^{-2}
P5	12.28	5.81×10^{-2}

4. Discussion

4.1. Chemical analysis

Chemical analyses of the glasses show that Na₂O losses during melting were low (0.2–1.8 mol%) except for glass P3, where ~4 mol% Na₂O was lost. CaO losses from all the glasses were negligible. The ZrO₂ content found by analysis was always lower than the calculated content, indicating that not all the ZrO₂ in the batch was dissolved in the glass. No undissolved ZrO₂ could be observed in glasses P1–P5 using either optical microscopy or X-ray diffraction. It is probable that the heavy ZrO₂ has settled at the bottom of the crucible where dissolution would be less likely to take place. It may then be

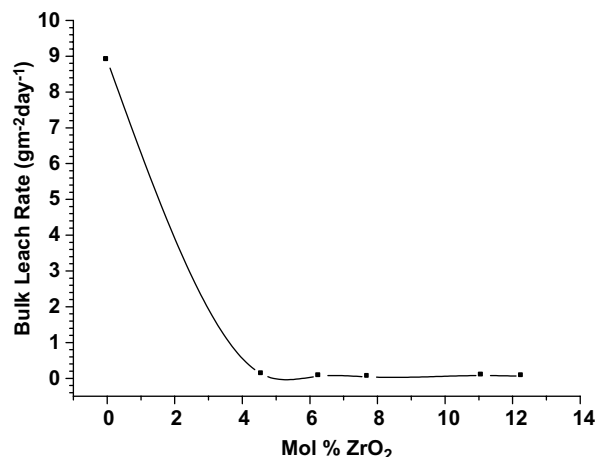


Fig. 8. Bulk leach rate (g m⁻² day⁻¹) vs. mol% ZrO₂ (analyzed).

retained in the crucible when the glass was poured [5]. Up to 12.3 mol% ZrO₂ could be incorporated into these glasses before unmelted ZrO₂ appeared in the glasses. This is equivalent to a ZrO₂ loading of 22.3 wt% for glass P5. This is comparable to the ZrO₂ weight loadings in high-zirconia borosilicate glasses [2,3] and the weight loadings in the silicate glasses melted by Wang et al. [5]. It is possible that by melting at a lower temperature (and therefore a higher viscosity) and stirring more ZrO₂ could be incorporated into the glass.

In all of the glasses, the SiO₂ content found by analysis was greater than the calculated content. This is a consequence of the loss of other components (Na₂O and ZrO₂). A slight rise in CaO concentration was found for two of the glasses (P4 and P7) and a slight rise in Na₂O concentration was found for glass P7. The expected rises in Na₂O concentration for glasses P1–P6 will have been offset by losses due to volatilization.

4.2. Differential thermal analysis

Differential thermal analysis results show that the T_g of the glasses increased with the addition of ZrO₂. The onset temperatures of the exothermic peaks for glasses P4 and P5 agree well with the temperatures at which the heat-treated glasses were found to crystallize. Therefore these peaks have been assigned as crystallization peaks. The slow rate at which these glasses crystallized at 1050 °C (crystallization had only just begun to occur after 12 h) also agrees well with the small size of the exothermic peaks. The area of the exothermic crystallization peak of a glass is dependent on the rate at which the glass devitrifies [12]. Binary glasses tend to devitrify rapidly and give large crystallization peaks whereas more complex glasses devitrify slowly and give smaller peaks. Sometimes, the rate of devitrification is so slow that no peak is observed. This may explain why no crystallization peaks were visible in the trace for glass P1.

The small area of the crystallization peaks for glasses P4 and P5 indicate that these glasses devitrify particularly slowly.

4.3. X-ray diffraction

The X-ray diffraction results show that the glasses are resistant to crystallization. Glasses P1 and P2 (with 4.6 and 6.2 mol% ZrO_2 as analyzed respectively) could not be made to crystallize at all. Glasses P3–P6 (7.7–14.7 mol% ZrO_2 as analyzed) began to crystallize at 1050 °C. Keldyshite and parakeldyshite were the first phases to crystallize at 1050 °C. Further crystallization of keldyshite and parakeldyshite occurred in all the glasses at 1150 °C.

When the results from optical microscopy and X-ray diffraction for glasses P3–P5 are compared, it is clear that the lath-like crystalline phase present in these glasses is a mixture of keldyshite and parakeldyshite. Due to its high reflectance and morphology, the dendritic phase observed as a minor phase in samples P3–P5 is thought to be monoclinic ZrO_2 [13]. There is an inconsistency between the optical microscopy results and the X-ray diffraction results for glass P6 heat-treated at 1150 °C for 12 h. The optical microscopy results suggest that dendritic ZrO_2 has crystallized (along with a lath-like phase), but the X-ray diffraction trace shows only keldyshite and parakeldyshite. The reason for this inconsistency is not known.

The resistance of glasses P1 and P2 (4.6 and 6.2 mol% ZrO_2 as analyzed respectively) to crystallization is comparable to that of Na_2O – ZrO_2 – SiO_2 glasses produced by Vargin and Kheifets [14]. They found that Na_2O – ZrO_2 – SiO_2 glasses with 5 mol% ZrO_2 did not crystallize in a gradient furnace within a temperature range of 800–1230 °C. As they increased the ZrO_2 content, the tendency of their glasses to crystallize increased, but they managed to produce glasses with up to 15 mol% ZrO_2 which would not crystallize in the gradient furnace within a temperature range of 800–1230 °C. Glasses P4 and P6 have compositions similar to the compositions of the two glasses of Vargin and Kheifets mentioned above. Both P4 and P6 begin to show crystallization at 1050 °C, with crystallization occurring throughout both glasses at 1150 °C. It would appear then that addition of CaO increases the tendency for a Na_2O – ZrO_2 – SiO_2 glass to crystallize. It must be borne in mind that Vargin and Kheifets did not specify the dwell time of their glasses in the furnace. They also did not specify how they determined that their glasses had begun to crystallize.

4.4. Chemical durability

From Fig. 8 it can be seen that addition of 4.6 mol% ZrO_2 caused a dramatic improvement in the resistance of the base glass P0 to attack by water. Further addi-

tions of ZrO_2 have no noticeable effect on bulk leach rate.

Comparisons between these results and the results of other workers are difficult due to the different testing methods used. The chemical durabilities of the ZrO_2 -containing borosilicate glasses produced by St-Pierre et al. were tested using the Soxhlet method on powder samples in distilled water at 100 °C for four days [3]. Their glasses had bulk leach rates varying from 7.78×10^{-2} to $9.42 \times 10^{-1} \text{ g m}^{-2} \text{ day}^{-1}$. The results in the present work are comparable to their results. It should be borne in mind that the test conditions in their work (glass powder and high temperature of leaching solution) would lead to higher bulk leach rates than the conditions in the present work (bulk sample and low temperature of leaching solution).

Luckscheiter and Nesovic measured the bulk leach rate of borosilicate glasses based on a German glass (GP-WAW) and the French glass R7T7 [15]. They used the Soxhlet method to leach cubic glass samples in distilled water at 100 °C for 30 days. Bulk leach rates were $4.8 \text{ g m}^{-2} \text{ day}^{-1}$ for GP-WAW and $2.5 \text{ g m}^{-2} \text{ day}^{-1}$ for R7T7. Delage and Dussossoy carried out Soxhlet leach tests in water at neutral pH on borosilicate glasses of the R7T7 composition [16]. For a sample leached at 90 °C for 9 days, the bulk mass leach rate was $1.0 \text{ g m}^{-2} \text{ day}^{-1}$. The bulk leach rates of the glasses in the present work are lower than these values. However, the Soxhlet leach tests in the present work were carried out at a lower temperature than the tests above, which would lead to lower values of bulk leach rate.

The behavior of the glasses in the present work are similar to those of Dimbleby et al. [7] for Na_2O – ZrO_2 – SiO_2 glasses in which Na_2O was replaced by ZrO_2 . Increasing the ZrO_2 content from 0.95 mol% to 2.67 mol% caused a dramatic improvement in the hydrolytic durability of their glass. Further additions of ZrO_2 resulted in a less marked improvement in durability. Additions of >5 mol% ZrO_2 had little effect on durability. In the results for the P series glasses, addition of 4.6 mol% ZrO_2 produced a marked improvement in the hydrolytic resistance of the glass. After this, further addition of ZrO_2 had little effect.

Comparison of the results with other work on Na_2O – CaO – ZrO_2 – SiO_2 glasses is complicated by the fact that both Navarro et al. [8] and Ohta and Suzuki [9] measured the amount of Na_2O extracted from their glasses, as opposed to the total weight loss. The results of both investigations showed that addition of ZrO_2 caused a continuous gradual reduction in the amount of Na_2O extracted from the glass. In the results of Ohta and Suzuki, the curve of the amount of Na_2O extracted vs. ZrO_2 content flattened out as ZrO_2 content increased. In the results for the present work, there was an initial sharp decrease in the bulk leach rate with addition of 4.6 mol% ZrO_2 . After this, further addition of ZrO_2 had little effect.

The thermodynamic behavior of ZrO_2 can be used to explain its effect on durability [17]. The hydration of ZrO_2 to $\text{Zr}(\text{OH})_2^{2+}$ or $\text{Zr}(\text{OH})_4$ is favorable at all pH values and is more favorable than the formation of ionic species such as ZrO^{2+} , Zr^{4+} and HZrO_3^- . Therefore, a hydrated ZrO_2 layer builds up on the surface of the glass and is thought to act as a barrier for diffusion of ionic species from the glass. Several workers have found evidence for the existence of hydrated ZrO_2 or ZrO_2 – SiO_2 layers on the surfaces of corroded ZrO_2 -containing glasses [18,19,8,20]. In our P series glasses, further additions above 4.6 mol% ZrO_2 had little effect on durability. It is possible that at a concentration of 4.6 mol%, a hydrated ZrO_2 layer has already formed on the surface of the glass and that further additions of ZrO_2 have no effect on this layer.

4.5. Suitability as a waste glass

The P series glasses show the ability to incorporate a respectable amount of ZrO_2 . Up to 22.3 wt% ZrO_2 could be incorporated into the glass, comparable to that incorporated into borosilicate and silicate glasses. By changing the melting conditions (lowering melting temperature and stirring the melt) it may be possible to increase the amount of ZrO_2 that can be incorporated into the glasses.

The P series glasses show good chemical durability and are resistant to crystallization, with crystallization beginning at temperatures in excess of 950 °C. However, the high melting temperatures (~300 °C higher than the ideal melting temperature of 1150 °C) are a serious drawback in view of the usage of such compositions for storage of radioactive wastes. The melting temperatures could be lowered by increasing the Na_2O content of the glasses. However, this would tend to reduce the glass transition and crystallization temperatures and also the chemical durability of the glasses.

5. Conclusions

- Glasses in the Na_2O – CaO – ZrO_2 – SiO_2 system containing up to 15.25 mol% ZrO_2 have been produced. Glasses were found to be free of crystalline devitrification products although glasses with ≥ 14.7 mol% ZrO_2 contained undissolved ZrO_2 from the batch. Glasses containing up to 6.3 mol% ZrO_2 had liquidus temperatures between 950 and 1050 °C. Glasses containing ≥ 7.7 mol% ZrO_2 had liquidus temperatures >1150 °C. All of the ZrO_2 -containing compositions produced viscous melts, requiring melting temperatures of ≥ 1450 °C in order to remove seed and produce a melt that could easily be cast. Volatilization losses of ≤ 4 mol% Na_2O were measured.
- Glass transition temperature and glass crystallization temperature increased with ZrO_2 content. The glass transition temperature increases from 509 °C (for a glass with 0 mol% ZrO_2) to 800 °C (for a glass with 14.7 mol% ZrO_2). Glasses with ≤ 6.3 mol% ZrO_2 do not show any crystallization peaks. Glasses with >6.3 mol% ZrO_2 show two crystallization peaks above 950 °C, with a separation of ~ 90 °C between the peaks. Glasses with >6.3 mol% ZrO_2 crystallize at temperatures ≥ 1050 °C. The first phases to crystallize are keldyshite and parakeldyshite at 1050 °C, along with a minor dendritic phase (possibly monoclinic ZrO_2). In glasses heat-treated at 1150 °C, the only phases present are keldyshite and parakeldyshite for glasses with ≤ 12.3 mol% ZrO_2 . For glass P6 (14.7 mol% ZrO_2), monoclinic ZrO_2 may also be present.
- Addition of up to 4.6 mol% ZrO_2 to the base Na_2O – CaO – SiO_2 composition causes a considerable increase in the hydrolytic resistance of the glasses. Further additions of ZrO_2 have relatively little effect.

Acknowledgements

This work was funded by the Engineering and Physical Sciences Research Council and British Glass Plc. under a CASE award. The authors would like to thank I. Watts and D. Haylock for operating the glass melting furnaces, C. Ahern at British Nuclear Fuels Ltd. UK Group for carrying out the Soxhlet leach tests, and G. Glossop for carrying out the chemical analyses.

References

- I.W. Donald, B.L. Metcalfe, R.N.J. Taylor, J. Mater. Sci. 32 (1997) 5851.
- D.A. McKeown, I.S. Muller, A.C. Buechele, I.L. Pegg, J. Non-Cryst. Solids 258 (1999) 98.
- J. St-Pierre, H.H. Tran, L. Zirkovsky, J. Nucl. Mater. 107 (1982) 286.
- D.A. McKeown, I.S. Muller, A.C. Buechele, I.L. Pegg, C.A. Kendziora, J. Non-Cryst. Solids 262 (2000) 126.
- T. Wang, R.J. Hand, P.F. James, C.R. Scales, Glass Technol. 43C (2002) 162.
- B.R.T. Frost, in: R.W. Cahn, P. Haasen, E.J. Krame (Eds.), Materials Science and Technology, Vol. 10B (Nuclear Materials), VCH, New York, 1994, p. 422.
- V. Dumbleby, S. English, E.M. Firth, F.W. Hodkin, W.E.S. Turner, J. Soc. Glass. Technol. 12 (1928) 52.
- F. Navarro, J. Ma, E.A. Mari, Glass '89, Proceedings of the 15th International Congress on Glass, Leningrad, vol. 2b, Plenum Press, New York, 1989, p. 113.
- H. Ohta, Y. Suzuki, Ceram. Bull. 57 (6) (1978) 601.
- K.A. Shahid, F.P. Glasser, Phys. Chem. Glass 12 (2) (1971) 50.
- C. Clark-Monks, J.M. Parker, Stones and Cord in Glass, Society of Glass Technology, 1980, p. 46.

- [12] F.W. Wilburn, J.B. Dawson, in: R.C. Mackenzie (Ed.), *Differential Thermal Analysis*, Vol. 2: Applications, Academic Press, London, 1972, p. 238.
- [13] C. Clark-Monks, J.M. Parker, *Stones and Cord in Glass*, Society of Glass Technology, 1980, p. 92.
- [14] V.V. Vargin, V.S. Kheifets, *Z. Prikl. Khim. (Sov. J. Phys. Chem.)* 37 (1964) 1369.
- [15] B. Luckscheiter, M. Nesovic, *Waste Manag.* 16 (7) (1996) 571.
- [16] F. Delage, J.L. Dussossoy, *Mater. Res. Soc. Sympos. Proc.* 212 (1991) 41.
- [17] A. Paul, *J. Mater. Sci.* 12 (1977) 2246.
- [18] M. Chakraborty, D. Das, S. Basu, A. Paul, *Int. J. Cement Compos.* 1 (3) (1979) 103.
- [19] A. Makishima, M. Tsutsumi, T. Shimohira, T. Nagata, *J. Am. Ceram. Soc.* 66 (8) (1983) C-139.
- [20] N. Koshizaki, *J. Mater. Sci. Lett.* 7 (1988) 1128.